

Network architecture: function $NN_\theta : \mathbb{R}^{\text{input-dim}} \rightarrow \mathbb{R}^{\text{output-dim}}$ parametrized by $\theta \equiv (W, b)$

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Loss function $L(\theta)$: is minimized during training.

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Optimization procedure (gradient descent): updates θ with $\nabla L(\theta)$; $\theta \leftarrow \theta - \eta \nabla L(\theta)$